

*****To Setup Jenkins on RHEL-9.8 and integrate to grafana & prometheus*****

Step 1: ping google and redhat on VM

```
#ping www.google.com
```

```
#ping www.redhat.com
```

Step 2: Take Snapshot

```
Name:-Before_install_jenkins_server
```

Step 3:- Set Hostname

```
#hostnamectl set-hostname jenkins.example.com
```

Step 4:- Mount CD/DVD to /mnt

```
#mount /dev/cdrom /mnt
```

```
#yum clean all
```

```
#yum repolist all
```

Note:-to enabled file based repo

Step 5:- Install Java (Jenkins requires Java)

```
#yum install java-21-openjdk -y
```

Verify Java:-

```
#java -version
```

```
[root@localhost opt]# java --version
openjdk 21.0.11 2026-04-21 LTS
OpenJDK Runtime Environment (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS, mixed mode, sharing)
[root@localhost opt]#
```

Step 6:- Enable Jenkins Repository

```
#wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

```
#rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

```
[root@localhost ~]#
[root@localhost ~]# wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2026-07-14 22:07:52-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.66.133, 151.101.194.133, 151.101.2.133, ...
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.66.133|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
--2026-07-14 22:07:54-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=====>]      267  --.-KB/s   in 0s

2026-07-14 22:07:54 (1.16 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@localhost ~]# rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

Step 7: Install Jenkins

```
#yum install jenkins -y
```

```
[root@localhost ~]# yum install jenkins -y
Updating Subscription Management repositories.
Jenkins-stable                               1.7 kB/s | 833
B      00:00
Jenkins-stable                               12 kB/s | 1.6
kB     00:00
Importing GPG key 0x14ABFC68:
  Userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From        : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Jenkins-stable                               5.8 kB/s | 2.5
kB      00:00
Dependencies resolved.
=====
1/1
Installed products updated.

Installed:
  jenkins-2.568.1-1.noarch

Complete!
```

Step 8: Start and Enable Jenkins

```
#systemctl start jenkins
```

```
#systemctl enable --now jenkins
```

#systemctl status jenkins

```
[root@localhost ~]# systemctl enable --now jenkins.service
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@localhost ~]# systemctl status jenkins.service
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 22:10:55 IST; 13s ago
     Main PID: 8635 (java)
    Tasks: 44 (limit: 10301)
   Memory: 424.2M (peak: 437.3M)
      CPU: 28.757s
   CGroup: /system.slice/jenkins.service
           └─8635 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war

Jul 14 22:10:44 localhost.localdomain jenkins[8635]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPa
Jul 14 22:10:44 localhost.localdomain jenkins[8635]: [LF]>
Jul 14 22:10:44 localhost.localdomain jenkins[8635]: [LF]> *****
*****
Jul 14 22:10:44 localhost.localdomain jenkins[8635]: [LF]> *****
*****
Jul 14 22:10:44 localhost.localdomain jenkins[8635]: [LF]> *****
*****
Jul 14 22:10:55 localhost.localdomain jenkins[8635]: 2026-07-14 16:40:55.252+0000 [id=32] INFO
jenkins.InitRead
Jul 14 22:10:55 localhost.localdomain jenkins[8635]: 2026-07-14 16:40:55.273+0000 [id=24] INFO
hudson.lifecycle
Jul 14 22:10:55 localhost.localdomain systemd[1]: Started Jenkins Continuous Integration Server.
Jul 14 22:10:59 localhost.localdomain jenkins[8635]: 2026-07-14 16:40:59.692+0000 [id=50] INFO
h.m.DownloadServ
Jul 14 22:10:59 localhost.localdomain jenkins[8635]: 2026-07-14 16:40:59.705+0000 [id=50] INFO
hudson.util.Retr
```

#jenkins --version

```
[root@localhost ~]# jenkins --version
2.568.1
```

Step 9: Adjust the Firewall

#firewall-cmd --permanent --add-port=8080/tcp

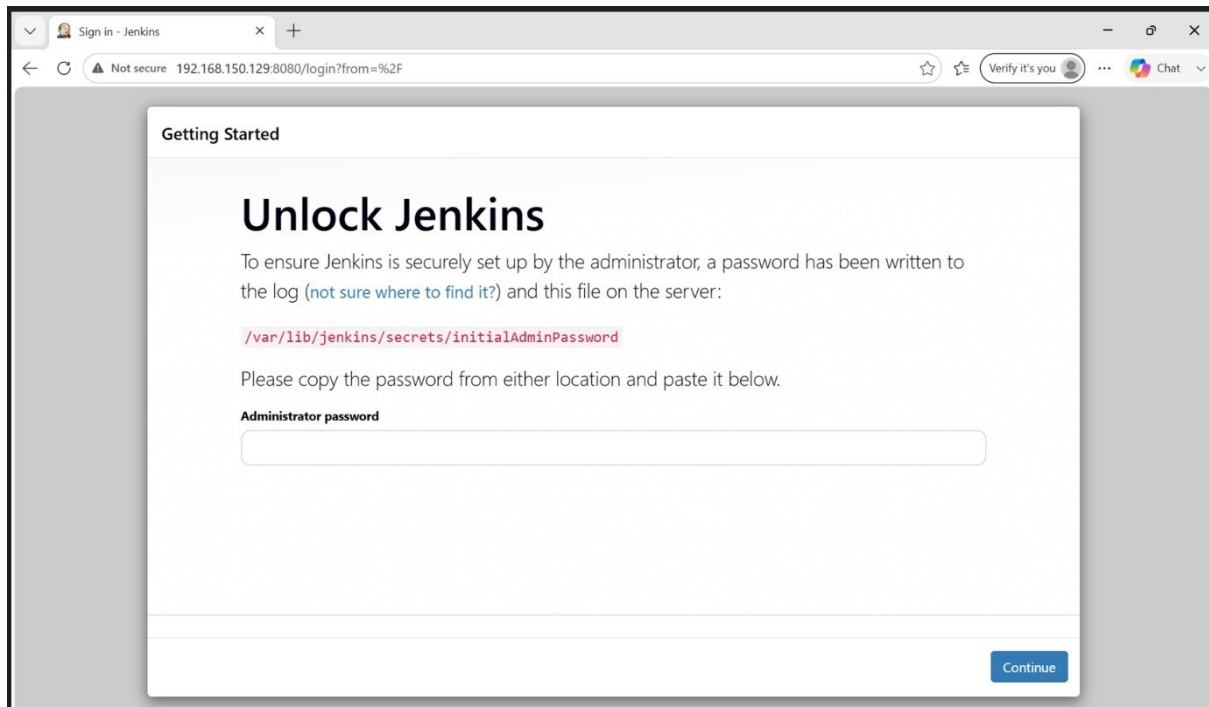
#firewall-cmd --reload

#firewall-cmd --list-all

```
[root@localhost ~]#
[root@localhost ~]# firewall-cmd --permanent --add-port=8080/tcp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
[root@localhost ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8080/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Step 10: Access Jenkins Web UI

`http://<your_server_ip>:8080`



Step 11: Get Initial Admin Password

```
#cat /var/lib/jenkins/secrets/initialAdminPassword
```

```
[root@localhost ~]# cat /var/lib/jenkins/secrets/initialAdminPassword  
e999cf9f4c344a6893cd0596c05af46b
```

Use this password to unlock Jenkins web interface.

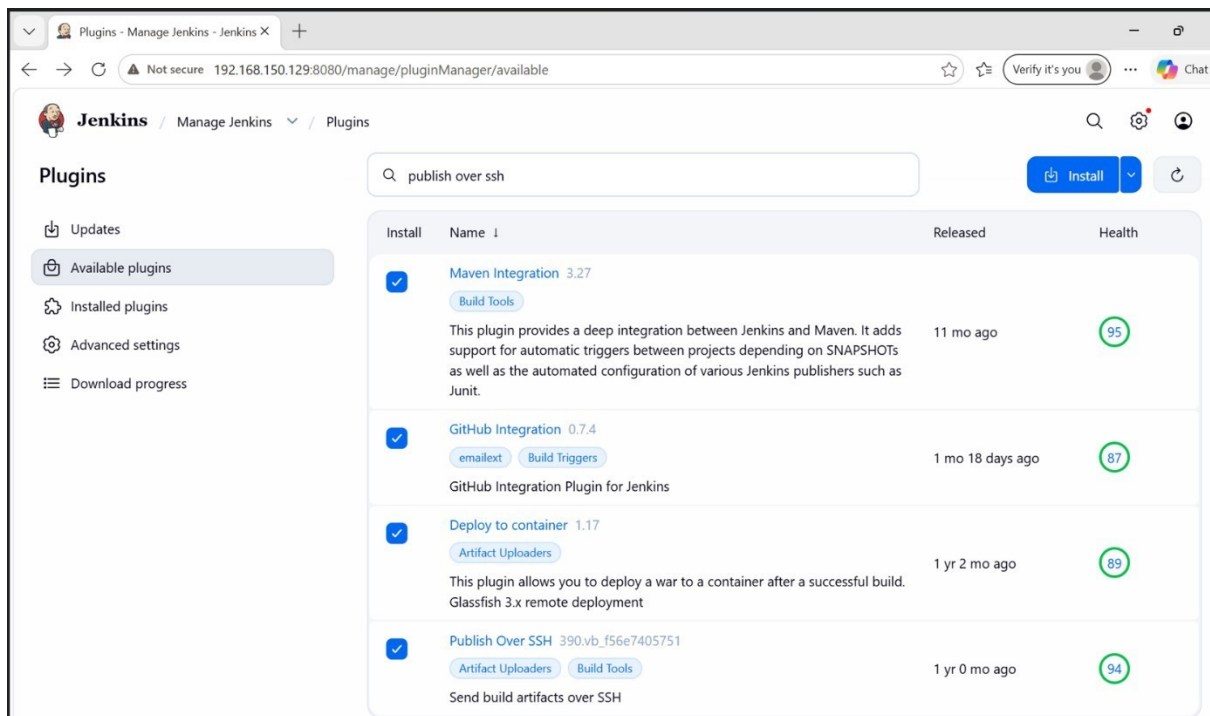
Step 12: Complete Jenkins Setup on Web UI

Install suggested plugins

Create admin user

Start using Jenkins

plugins install:-GitHub Integration,maven integration,Deploy to container,Publish Over SSH



Step 13:-Install maven on jenkins server

go to browser and type apache maven download

and select tar.gz file {<https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz>}

```
#cd /opt
```

```
#wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
```

```
#ls
```

```
#tar -xvzf apache-maven-3.9.16-bin.tar.gz
```

```
#ls
```

```
#mv apache-maven-3.9.16 maven
```

```
#ls
```

```
[root@localhost opt]#
[root@localhost opt]# wget https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
--2026-07-14 22:15:00-- https://dlcdn.apache.org/maven/maven-3/3.9.16/binaries/apache-maven-3.9.16-bin.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9278065 (8.8M) [application/x-gzip]
Saving to: 'apache-maven-3.9.16-bin.tar.gz'

apache-maven-3.9.16-bin.tar.gz 100%[=====>] 8.85M 1.95MB/s
in 4.3s

2026-07-14 22:15:04 (2.07 MB/s) - 'apache-maven-3.9.16-bin.tar.gz' saved [9278065/9278065]

[root@localhost opt]#
[root@localhost opt]# ls
apache-maven-3.9.16-bin.tar.gz
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]# tar -xvzf apache-maven-3.9.16-bin.tar.gz
apache-maven-3.9.16/README.txt
apache-maven-3.9.16/LICENSE
apache-maven-3.9.16/NOTICE
apache-maven-3.9.16/lib/
apache-maven-3.9.16/lib/aopalliance.license
apache-maven-3.9.16/lib/asm.license
```

```
[root@localhost opt]# ls
apache-maven-3.9.16-bin.tar.gz maven
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]# rm -rvf apache-maven-3.9.16-bin.tar.gz
removed 'apache-maven-3.9.16-bin.tar.gz'
```

Step 14:-Now to find java path

```
#find /usr/lib/jvm/java-21* | head -n 3
```

```
/usr/lib/jvm/java-21-openjdk-21.0.8.0.9-1.el9.x86_64
```

```
[root@localhost opt]# find /usr/lib/jvm/java-21* | head -n 3
/usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64
/usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64/bin
/usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64/bin/alt-java
```

Step 15:- Now to open .bashrc and .bash_profile file

```
#vim .bash_profile
```

```
JAVA_HOME=/usr/lib/jvm/java-21-openjdk-21.0.8.0.9-1.el9.x86_64
```

```
M2_HOME=/opt/maven
```

```
M2=/opt/maven/bin
```

```
PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2:$M2_HOME
```

```
export PATH
```

```
:wq!
```

```
#vim .bashrc

JAVA_HOME=/usr/lib/jvm/java-21-openjdk-21.0.8.0.9-1.el9.x86_64

M2_HOME=/opt/maven

M2=/opt/maven/bin

PATH=$PATH:$HOME/bin:$JAVA_HOME:$M2:$M2_HOME

export PATH
```

:wq!

```
#source .bashrc

#source .bash_profile

#echo $JAVA_HOME

#echo $M2_HOME

#echo $M2

#java -version

#mvn -version
```

```
[root@localhost opt]# java --version
openjdk 21.0.11 2026-04-21 LTS
OpenJDK Runtime Environment (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-21.0.11.0.10-1) (build 21.0.11+10-LTS, mixed m
ode, sharing)
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]# mvn --version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: /opt/maven
Java version: 21.0.11, vendor: Red Hat, Inc., runtime: /usr/lib/jvm/java-21-open
jdk-21.0.11.0.10-2.el9.x86_64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "5.14.0-687.17.1.el9_8.x86_64", arch: "amd64", family
: "unix"
[root@localhost opt]#
[root@localhost opt]# echo $JAVA_HOME
/usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64
[root@localhost opt]#
[root@localhost opt]# echo $M2_HOME
/opt/maven
```

Step 16:- {to create maven directory structure}

```
#mvn archetype:generate | grep maven-archetype-webapp {choose sample
maven webapp project which no is 2267}
```

```
[root@localhost opt]# mvn archetype:generate | grep maven-archetype-webapp
741: remote -> com.haoxuer.maven.archetype:maven-archetype-webapp (a simple maven archetype)
904: remote -> com.lodsve:lodsve-maven-archetype-webapp (Lodsve Maven Archetype Webapp)
2367: remote -> org.apache.maven.archetypes:maven-archetype-webapp (An archetype which contains a sample Maven Webapp project.)
2635: remote -> org.bytesizebook.com.guide.boot:maven-archetype-webapp (An archetype to create the starting code for the first three chapters of Guide to Web Development with Java, 2nd edition.)
46
```

#mvn archetype:generate

Choose a number: 8: 8

Define value for property 'groupId': in.radical.webapp

Define value for property 'artifactId': FormFillApp

```
Choose a number or apply filter (format: [groupId:]artifactId, case sensitive contains): 2362: 2367
Choose org.apache.maven.archetypes:maven-archetype-webapp version:
1: 1.0-alpha-1
2: 1.0-alpha-2
3: 1.0-alpha-3
4: 1.0-alpha-4
5: 1.0
6: 1.3
7: 1.4
8: 1.5
Choose a number: 8: 8
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.5/maven-archetype-webapp-1.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/archetypes/maven-archetype-webapp/1.5/maven-archetype-webapp-1.5.jar (9.9 kB at 5.1 kB/s)
Define value for property 'groupId': in.radical.webapp
Define value for property 'artifactId': FormFillApp
Define value for property 'version' 1.0-SNAPSHOT:
Define value for property 'package' in.radical.webapp:
Confirm properties configuration:
groupId: in.radical.webapp
artifactId: FormFillApp
version: 1.0-SNAPSHOT
package: in.radical.webapp
Y:
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.5
[INFO] -----
[INFO] Parameter: groupId, Value: in.radical.webapp
[INFO] Parameter: artifactId, Value: FormFillApp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: in.radical.webapp
[INFO] Parameter: packageInPathFormat, Value: in/radical/webapp
[INFO] Parameter: package, Value: in.radical.webapp
[INFO] Parameter: groupId, Value: in.radical.webapp
[INFO] Parameter: artifactId, Value: FormFillApp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[WARNING] CP Don't override file /opt/FormFillApp/src/main/webapp
[WARNING] CP Don't override file /opt/FormFillApp/.mvn
[INFO] Project created from Archetype in dir: /opt/FormFillApp
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 46.866 s
[INFO] Finished at: 2026-07-14T22:20:43+05:30
[INFO] -----
[root@localhost opt]#
```

#tree FormFillApp/

#cd FormFillApp

#vim src/main/webapp/index.jsp

<form action="action_page.php">

<div class="container">

<h1>Radical Student Registration!</h1>

<p>Please fill in this form!</p>

<hr>

<label for="email">Email</label>

<input type="text" placeholder="Enter Email" name="email" id="email" required>

<label for="psw">Password</label>

<input type="password" placeholder="Enter Password" name="psw" id="psw" required>

<label for="psw-repeat">Repeat Password</label>

<input type="password" placeholder="Repeat Password" name="psw-repeat" id="psw-repeat" required>

<hr>

<p>By creating an account you agree to our Terms & Privacy.</p>

<button type="submit" class="registerbtn">Register</button>

</div>

<div class="container signin">

<p>Already have an account? Sign in.</p>

</div>

</form>

:wq!

Step 17:- Now to install git on jenkins server

```
#yum install git -y
```

```
#git init
```

```
#git config --global user.name "Sai"
```

```
#git config --global user.email "sai3cs@gmail.com"
```

```
#git config --list
```

```
#git add .
```

```
#git status
```

```
#git commit -m "myfirstcode"
```

```
[root@localhost FormFillApp]# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /opt/FormFillApp/.git/
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git config --global user.name "sd"
[root@localhost FormFillApp]# git config --global user.email "sd@gmail.com"
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git config --list
user.name=sd
user.email=sd@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
[root@localhost FormFillApp]# git add .
[root@localhost FormFillApp]#
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .mvn/jvm.config
        new file:   .mvn/maven.config
        new file:   pom.xml
        new file:   src/main/webapp/WEB-INF/web.xml
        new file:   src/main/webapp/index.jsp

[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git commit -m "my first code"
[master (root-commit) 4a4c035] my first code
 5 files changed, 102 insertions(+)
 create mode 100644 .mvn/jvm.config
 create mode 100644 .mvn/maven.config
 create mode 100644 pom.xml
 create mode 100644 src/main/webapp/WEB-INF/web.xml
 create mode 100644 src/main/webapp/index.jsp
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git log
commit 4a4c03557954be0e6cb3f0ba84f95a4ae575dda6 (HEAD -> master)
Author: sd <sd@gmail.com>
Date: Tue Jul 14 22:25:45 2026 +0530

    my first code
```

Now we create central repo with same name on github account

```
#git remote add origin https://github.com/dpatel136/FormFillApp.git
```

```
#git branch -M main
```

```
#git push -u origin main
```

```

[root@localhost FormFillApp]# git remote add origin https://github.com/SMSiddhesh/FormFillApp.git
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git branch -M main
[root@localhost FormFillApp]#
[root@localhost FormFillApp]# git push -u origin main
Username for 'https://github.com': sd
Password for 'https://sd@github.com':
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (11/11), 1.80 KiB | 230.00 KiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To https://github.com/SMSiddhesh/FormFillApp.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

Step 18:- Now go back to jenkins dashboard

go to manage jenkins :-inside tools

Add JDK

Add git

Add maven

apply and save

Maven installations ^
Edited

+ Add Maven

Maven

Name

maven

MAVEN_HOME

/opt/maven

☐ Install automatically ?

JDK installations ^
Edited

+ Add JDK

JDK

Name

java-21

JAVA_HOME

/usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64

☒ /usr/lib/jvm/java-21-openjdk-21.0.11.0.10-2.el9.x86_64 doesn't look like a JDK directory

☐ Install automatically ?

Step 19:- to create 1 jenkins job

job name:-My_First_Maven_Project

git:- to paste github account FormFillApp repo url

branch:-main

Goals and options:-clean install

apply and save and build a job

The screenshot shows the Jenkins job configuration page for a job named 'My_First_Maven_Project'. The 'Git' provider is selected. The 'Repository URL' is set to 'https://github.com/SMSiddhesh/FormFillApp.git'. The 'Root POM' is set to 'pom.xml'. The 'Goals and options' are set to 'clean install'. The 'Branches to build' section shows a 'Branch Specifier (blank for \'any\')' set to '*/main'.

Git ?

Repositories ?

Repository URL ?

https://github.com/SMSiddhesh/FormFillApp.git

Build

Root POM ?

pom.xml

Goals and options ?

clean install

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Now Install Node Exporter on Jenkins Server

Step 1:- Download Node Exporter

```
#cd /opt
```


User=node_exporter
Group=node_exporter
Type=simple
ExecStart=/opt/node_exporter/node_exporter

[Install]

WantedBy=multi-user.target

:wq!

Step 5:- Reload Systemd

```
#systemctl daemon-reload
```

Step 6:- Enable Service

```
#systemctl enable --now node_exporter
```

Step 7:- Check Status

```
#systemctl status node_exporter
```

```
[root@localhost opt]#  
[root@localhost opt]# systemctl daemon-reload  
[root@localhost opt]#  
[root@localhost opt]#  
[root@localhost opt]# systemctl enable --now node_exporter.service  
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.  
[root@localhost opt]#  
[root@localhost opt]# systemctl status node_exporter.service  
● node_exporter.service - Prometheus Node Exporter  
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-07-14 22:48:10 IST; 12s ago  
 Main PID: 12380 (node_exporter)  
    Tasks: 5 (limit: 10301)  
  Memory: 5.1M (peak: 5.4M)  
     CPU: 21ms  
    CGroup: /system.slice/node_exporter.service  
            └─12380 /opt/node_exporter/node_exporter
```

Step 8:- Open Firewall

```
#firewall-cmd --permanent --add-port=9100/tcp
```

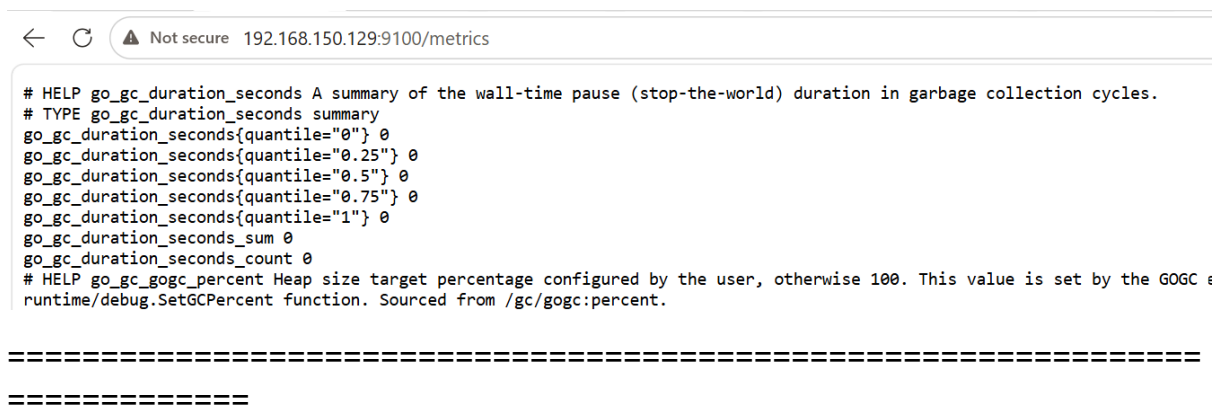
```
#firewall-cmd --reload
```

#firewall-cmd --list-all

```
[root@localhost opt]# firewall-cmd --permanent --add-port=9100/tcp
success
[root@localhost opt]#
[root@localhost opt]# firewall-cmd --reload
success
[root@localhost opt]#
[root@localhost opt]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8080/tcp 9100/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Step 9:- Verify Open in browser:

http://<jenkins-serverip>:9100/metrics



← ↻ ⚠ Not secure 192.168.150.129:9100/metrics

```
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC e
runtime/debug.SetGCPercent function. Sourced from /gc/gogc:percent.
```

=====

=====

Now to install Apache Exporter

Step 1:- Download Apache Exporter

```
#cd /opt
```

```
#wget
```

```
https://github.com/Lusitaniae/apache_exporter/releases/download/v1.0.9/apache_ex
porter-1.0.9.linux-amd64.tar.gz
```

Step 2:- Extract the files

```
#ls
```

#ls

Type=simple

```
ExecStart=/opt/apache_exporter/apache_exporter \
--scrape_uri=http://localhost/server-status?auto
```

Restart=always

[Install]

WantedBy=multi-user.target

:wq!

Step 6:- Reload Systemd

```
#systemctl daemon-reload
```

Step 7:- Enable Service

```
#systemctl enable --now apache_exporter
```

Step 8:- Check Status

```
#systemctl status apache_exporter
```

```
[root@localhost opt]# systemctl daemon-reload
[root@localhost opt]#
[root@localhost opt]# systemctl enable --now apache_exporter.service
Created symlink /etc/systemd/system/multi-user.target.wants/apache_exporter.service → /etc/systemd/system/apache_exporter.service.
[root@localhost opt]#
[root@localhost opt]#
[root@localhost opt]# systemctl status apache_exporter.service
● apache_exporter.service - Prometheus Apache Exporter
   Loaded: loaded (/etc/systemd/system/apache_exporter.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 22:56:42 IST; 10s ago
     Main PID: 12762 (apache_exporter)
        Tasks: 7 (limit: 10301)
       Memory: 4.6M (peak: 4.8M)
          CPU: 12ms
      CGroup: /system.slice/apache_exporter.service
              └─12762 /opt/apache_exporter/apache_exporter "--scrape_uri=http://localhost/server-status?auto"
```

Step 9:- Open Firewall

```
#firewall-cmd --permanent --add-port=9117/tcp
```

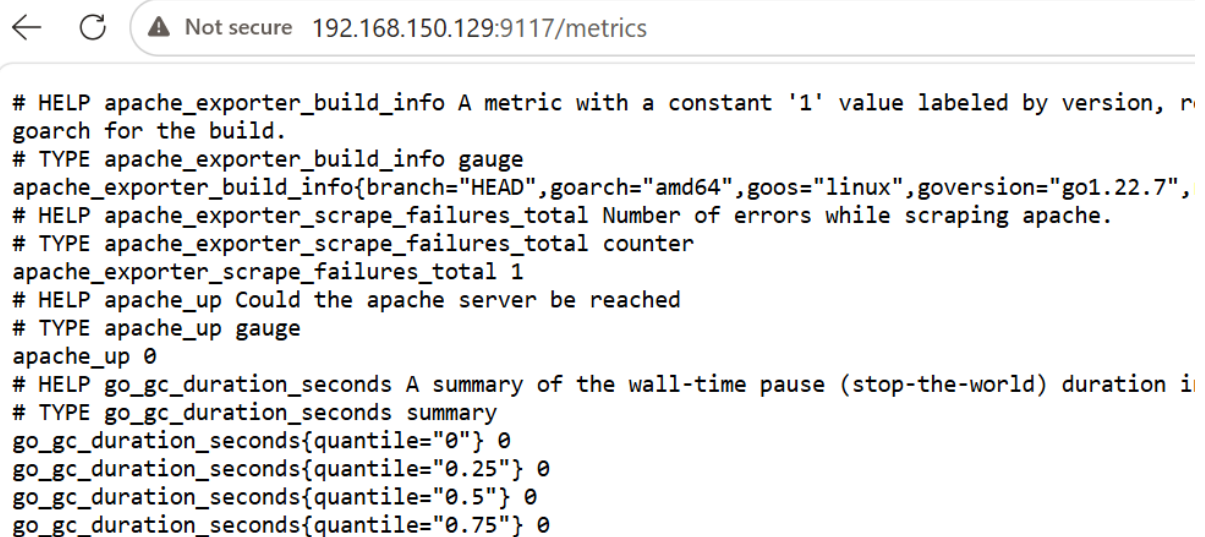
```
#firewall-cmd --reload
```

```
#firewall-cmd --list-all
```

```
[root@localhost opt]# firewall-cmd --reload
success
[root@localhost opt]#
[root@localhost opt]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8080/tcp 9100/tcp 9117/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

Step 9:- Verify Open in browser:

<http://<jenkins-serverip>:9117/metrics>



```
# HELP apache_exporter_build_info A metric with a constant '1' value labeled by version, runtime, and goarch for the build.
# TYPE apache_exporter_build_info gauge
apache_exporter_build_info{branch="HEAD",goarch="amd64",goos="linux",goversion="go1.22.7",} 1
# HELP apache_exporter_scrape_failures_total Number of errors while scraping apache.
# TYPE apache_exporter_scrape_failures_total counter
apache_exporter_scrape_failures_total 1
# HELP apache_up Could the apache server be reached
# TYPE apache_up gauge
apache_up 0
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in seconds.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
```

****Install Grafana on RHEL-9.8 Created By Dinesh Patel****

Step 1: ping google and redhat on VM

```
#ping www.google.com
```

```
#ping www.redhat.com
```

Step 2: Take Snapshot

Name:-Before_install_Grafana_server

Step 3:- Set Hostname

```
#hostnamectl set-hostname grafana.radical.com
```

Step 4:- Mount CD/DVD to /mnt

```
#mount /dev/cdrom /mnt
```

```
#yum clean all
```

```
#yum repolist all
```

Note:-to enabled file based repo

Step 5:- Add the Grafana RPM Repository

```
#wget -q -O /tmp/grafana_gpg.key https://rpm.grafana.com/gpg.key
```

```
#rpm --import /tmp/grafana_gpg.key
```

```
#vi /etc/yum.repos.d/grafana.repo
```

[grafana]

name=Grafana

baseurl=https://rpm.grafana.com

repo_gpgcheck=1

enabled=1

gpgcheck=1

gpgkey=https://rpm.grafana.com/gpg.key

sslverify=1

sslcert=/etc/pki/tls/certs/ca-bundle.crt

:wq! ---->{forcefully save & quite}

```
[root@grafana ~]#  
[root@grafana ~]# wget -q -O /tmp/grafana_gpg.key https://rpm.grafana.com/gpg.key  
[root@grafana ~]# rpm --import /tmp/grafana_gpg.key  
[root@grafana ~]#  
[root@grafana ~]# vi /etc/yum.repos.d/grafana.repo
```

Step 6:- Install Grafana

Once the repo is configured, install Grafana

#yum install grafana -y

#grafana server --version

```
[root@grafana ~]# grafana --version  
grafana version 13.1.0
```

Step 7:- Start and Enable Grafana Service

#systemctl start grafana-server.service

#systemctl enable --now grafana-server.service

#systemctl status grafana-server.service

```
[root@grafana ~]# systemctl enable --now grafana-server.service  
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.  
[root@grafana ~]#  
[root@grafana ~]#  
[root@grafana ~]# systemctl status grafana-server.service  
● grafana-server.service - Grafana instance  
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-07-14 23:11:54 IST; 14s ago  
     Docs: http://docs.grafana.org  
    Main PID: 5332 (grafana)  
      Tasks: 34 (limit: 10301)  
    Memory: 313.1M (peak: 351.1M)  
       CPU: 3.068s  
    CGroup: /system.slice/grafana-server.service  
            └─5332 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/var/run/grafana/grafana  
            └─5381 /var/lib/grafana/plugins/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64  
            └─5388 /var/lib/grafana/plugins/zipkin/gpx_grafana_zipkin-datasource_linux_amd64
```

Step 8:- Open the grafana port in firewall

#firewall-cmd --permanent --add-port=3000/tcp

#firewall-cmd --reload

#firewall-cmd --list-all

#netstat -antlpu | grep 3000

#ss -antlpu | grep 3000

```
[root@grafana ~]# firewall-cmd --permanent --add-port=3000/tcp
success
[root@grafana ~]# firewall-cmd --reload
success
[root@grafana ~]#
[root@grafana ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 8080/tcp 3000/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

```
[root@grafana ~]# netstat -antlpu |grep 3000
tcp6      0      0 :::3000          :::*             LISTEN     5332/grafana
[root@grafana ~]#
```

Step 9:- Verify it's working

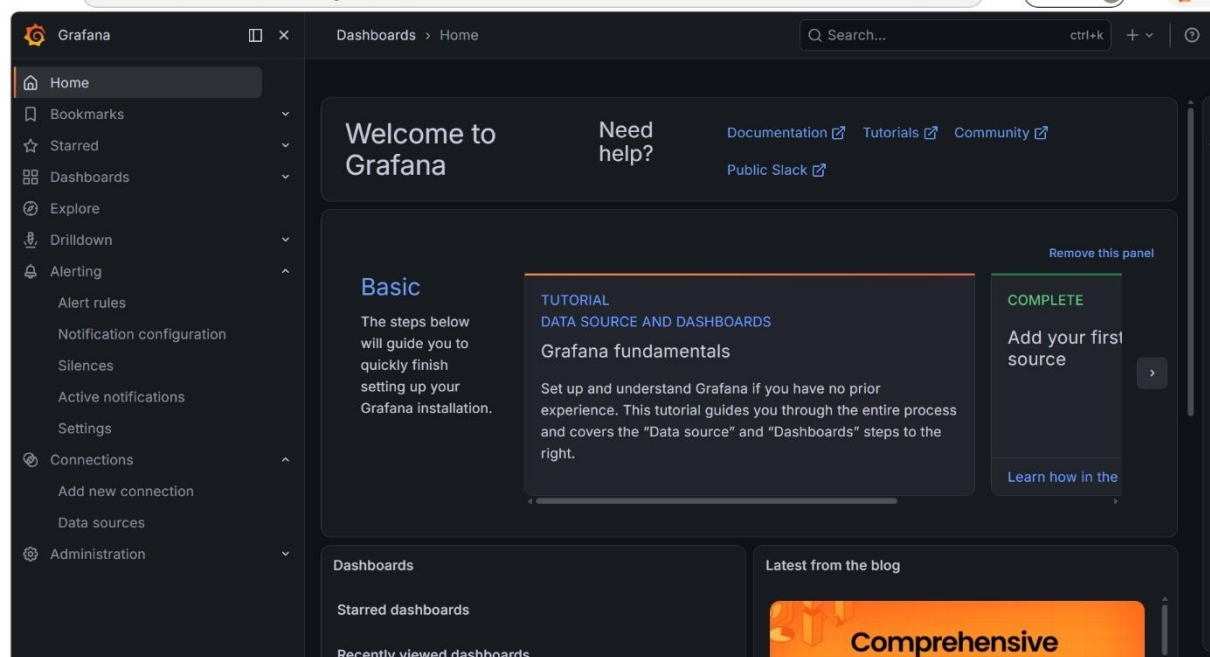
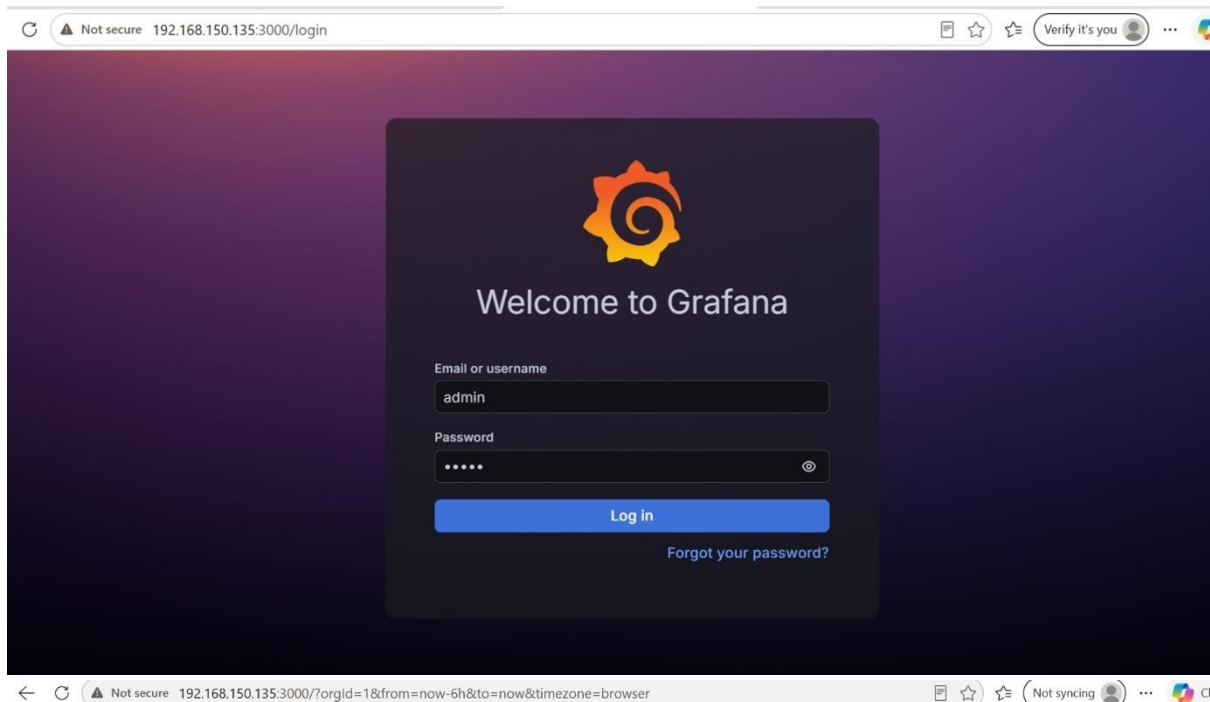
Open any browser and type below url

http://<your-server-ip>:3000

The default login credentials are:-

Username: admin

Password: admin



****Install Prometheus on RHEL-9.8 Created By Dinesh Patel****

Step 1: ping google and redhat on VM

```
#ping www.google.com
```

```
#ping www.redhat.com
```

Step 2: Take Snapshot

Name:-Before_install_prometheus_server

Step 3:- Set Hostname

```
#hostnamectl set-hostname prometheus.radical.com
```

Step 4:- Mount CD/DVD to /mnt

```
#mount /dev/cdrom /mnt
```

```
#yum clean all
```

```
#yum repolist all
```

Note:-to enabled file based repo

Step 5:- Create a dedicated service account

```
#useradd --no-create-home --shell /bin/false prometheus
```

```
[root@prometheus ~]# useradd --no-create-home --shell /bin/false prometheus  
[root@prometheus ~]#
```

Step 6:- Download Prometheus

Go to prometheus.io/download and find the latest stable version number.

```
#cd /opt
```

<https://github.com/prometheus/prometheus/releases/download/v3.12.0/prometheus-3.12.0.linux-amd64.tar.gz>

```
#ls
```

#ls

#ls

```
#cd prometheus
```

```
[root@prometheus ~]# cd /opt/[root@prometheus ~]# [root@prometheus ~]# wget https://github.com/prometheus/prometheus/releases/download/v3.12.0/prometheus-3.12.0.linux-amd64.tar.gz 2026-07-14 22:12:52... https://github.com/prometheus/prometheus/releases/download/v3.12.0/prometheus-3.12.0.linux-amd64.tar.gz Resolving github.com (github.com)... 98.207.73.82... Connecting to github.com (github.com)... 98.207.73.82:443... connected. HTTP request sent, waiting response... 202 Found Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838912/cdd3d68-57c9-4ccc-9c10-e698741756e7?rs=2018-11-09&sr=b&sp=http&se=2026-07-14T18JAS1N3A31Z&rscd=attachment%3B%2Ffilesname%3Dprometheus-3.12.0.linux-amd64.tar.gz&src=application%2Foctet-stream&skid=96c2d4f6-5711-43d1-add-ab047aa7ab0&skt=3986a654-997b-47e0-b1220-951b08b046e&skt=2026-07-14T17JAS08M47Z&sk=2026-07-14T18JAS1N3A31Z&sk=bskv=2018-11-09&sig=rHhGxHj2iSfLwXoSk9ehN7lU8mJz2FqZ3dACShK9K3dKyteYJe0kAM1QJVQ1CLCh3bc0c10Jzu1Iln39r_eyPcPM01nAx8MbduLvZ9T1vWVKKyjcsmZwF2zi3lc3mdImw20lvauVdnxmCmwentBrQuvZ9T1vWta2VijoiAzV3MS1mV4CCIMOfC4MDAINTErHWtbkmJJoXgZGDMDuXNTeYLcVxKnho10CwVdZMcZcvbnHv5t6B9HLmVucsdLuzG93cyZuZiqRq_TenZ07-RZF3RAP7VZ0ZIC9JRHTm_KVTUCmWwwowresponcent-disposition=attachment%3B%2Ffilesname%3Dprometheus-3.12.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream%5Bfollowing%5D -2026-07-14 22:12:53... https://release-assets.githubusercontent.com/github-production-release-asset/6838912/cdd3d68-57c9-4ccc-9c10-e698741756e7?rs=2018-11-09&sr=b&sp=http&se=2026-07-14T18JAS1N3A31Z&rscd=attachment%3B%2Ffilesname%3Dprometheus-3.12.0.linux-amd64.tar.gz&src=application%2Foctet-stream&skid=96c2d4f6-5711-43d1-add-ab047aa7ab0&skt=3986a654-997b-47e0-b1220-951b08b046e&skt=2026-07-14T17JAS08M47Z&sk=2026-07-14T18JAS1N3A31Z&sk=bskv=2018-11-09&sig=rHhGxHj2iSfLwXoSk9ehN7lU8mJz2FqZ3dACShK9K3dKyteYJe0kAM1QJVQ1CLCh3bc0c10Jzu1Iln39r_eyPcPM01nAx8MbduLvZ9T1vWVKKyjcsmZwF2zi3lc3mdImw20lvauVdnxmCmwentBrQuvZ9T1vWta2VijoiAzV3MS1mV4CCIMOfC4MDAINTErHWtbkmJJoXgZGDMDuXNTeYLcVxKnho10CwVdZMcZcvbnHv5t6B9HLmVucsdLuzG93cyZuZiqRq_TenZ07-RZF3RAP7VZ0ZIC9JRHTm_KVTUCmWwwowresponcent-disposition=attachment%3B%2Ffilesname%3Dprometheus-3.12.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream%5Bfollowing%5D -Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.108.133, ... Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133:443... connected. HTTP request sent, waiting response... 200 OK Length: 15277394 (44M) [application/octet-stream] Saving to: 'prometheus-3.12.0.linux-amd64.tar.gz' prometheus-3.12.0.linux-amd64.t 100%[#####] 145.70M 4.35MB/s in 21s 2026-07-14 22:12:14 (6.98 MB/s) - 'prometheus-3.12.0.linux-amd64.tar.gz' saved [15277394/15277394] [root@prometheus ~]# [root@prometheus ~]# [root@prometheus ~]# tar -xzf prometheus-3.12.0.linux-amd64.tar.gz [root@prometheus ~]# ls [root@prometheus ~]# cd prometheus-3.12.0.linux-amd64/ [root@prometheus ~]# ls [root@prometheus ~]# cd LICENSE [root@prometheus ~]# cd NOTICE [root@prometheus ~]# cd prometheus.yml [root@prometheus ~]# cd prometheus-3.12.0.linux-amd64/ [root@prometheus ~]# cd prometheus [root@prometheus ~]# [root@prometheus ~]# [root@prometheus ~]# ls [root@prometheus ~]# mv prometheus-3.12.0.linux-amd64 prometheus [root@prometheus ~]# [root@prometheus ~]# [root@prometheus ~]# ls [root@prometheus ~]# mv prometheus-3.12.0.linux-amd64.tar.gz
```

Create the standard directories:

```
#mkdir -p /var/lib/prometheus
```

```
[root@prometheus prometheus]# mkdir -p /etc/prometheus
[root@prometheus prometheus]# mkdir -p /var/lib/prometheus
[root@prometheus prometheus]#
```

Step 9:- Copy the

```
#cp prometheus promtool /usr/local/bin/
```

```
[root@prometheus prometheus]# cp prometheus promtool /usr/local/bin/
[root@prometheus prometheus]#
```

Step 10:- Copy the config and web UI assets

```
#cp prometheus.yml /etc/prometheus/
```

Step 11:- Set file ownership

```
#chown -R prometheus:prometheus /etc/prometheus
```

```
#chown -R prometheus:prometheus /var/lib/prometheus
```

```
#chown prometheus:prometheus /usr/local/bin/prometheus
```

```
#chown prometheus:prometheus /usr/local/bin/promtool
```

```
[root@prometheus prometheus]#  
[root@prometheus prometheus]# chown -R prometheus:prometheus /etc/prometheus  
[root@prometheus prometheus]#  
[root@prometheus prometheus]# chown -R prometheus:prometheus /var/lib/prometheus  
[root@prometheus prometheus]#  
[root@prometheus prometheus]# chown prometheus:prometheus /usr/local/bin/prometheus  
[root@prometheus prometheus]#  
[root@prometheus prometheus]# chown prometheus:prometheus /usr/local/bin/promtool  
[root@prometheus prometheus]#
```

Note:- Without this step, Prometheus will start as the prometheus user but be unable to read its config or write to the data directory.

You'll get permission denied at startup — confusing if you don't know where to look.

Step 12:- Create the systemd service file

```
#vi /etc/systemd/system/prometheus.service
```

[Unit]

Description=Prometheus Monitoring System

Wants=network-online.target

After=network-online.target

[Service]

User=prometheus

Group=prometheus

Type=simple

Restart=on-failure

ExecStart=/usr/local/bin/prometheus \

--config.file=/etc/prometheus/prometheus.yml \

--storage.tsdb.path=/var/lib/prometheus/ \

```
--web.console.templates=/etc/prometheus/consoles \  
--web.console.libraries=/etc/prometheus/console_libraries
```

[Install]

WantedBy=multi-user.target

:wq! ---->{forcefully save & quite}

Step 13:- Start and enable the service

```
#systemctl daemon-reload  
  
#systemctl start prometheus  
  
#systemctl enable --now prometheus  
  
#systemctl status prometheus
```

```
[root@prometheus prometheus]# systemctl enable --now prometheus.service  
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.  
[root@prometheus prometheus]#  
[root@prometheus prometheus]# systemctl status prometheus.service  
● prometheus.service - Prometheus Monitoring System  
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: disabled)  
   Active: active (running) since Tue 2026-07-14 23:26:43 IST; 11s ago  
 Main PID: 4881 (prometheus)  
    Tasks: 8 (limit: 10301)  
  Memory: 108.8M (peak: 114.3M)  
     CPU: 181ms  
   CGroup: /system.slice/prometheus.service  
           └─4881 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus
```

```
#prometheus --version
```

```
[root@prometheus opt]# prometheus --version  
prometheus, version 3.12.0 (branch: HEAD, revision: 9f27dffc1f93ca23287972f632025879f2d1c658)  
  build user:      root@2f9dfa1df447  
  build date:      20260528-15:49:37  
  go version:      go1.26.3  
  platform:        linux/amd64  
  tags:            netgo,builtinassets
```

Step 14:- Open the prometheus port in firewall

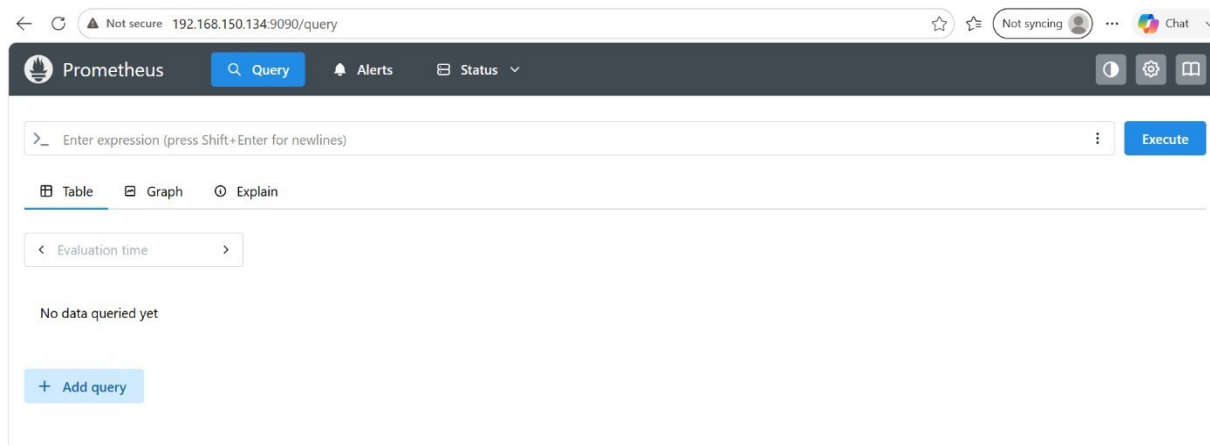
```
#firewall-cmd --permanent --add-port=9090/tcp  
  
#firewall-cmd --reload  
  
#firewall-cmd --list-all  
  
#netstat -antlpu | grep 9090  
  
#ss -antlpu | grep 9090
```

```
[root@prometheus prometheus]# firewall-cmd --permanent --add-port=9090/tcp
success
[root@prometheus prometheus]#
[root@prometheus prometheus]# firewall-cmd --reload
success
[root@prometheus prometheus]# \
> ^C
[root@prometheus prometheus]#
[root@prometheus prometheus]# firewall-cmd --list-all
public (active)
target: default
icmp-block-inversion: no
interfaces: ens160
sources:
services: cockpit dhcpv6-client ssh
ports: 8080/tcp 9090/tcp
protocols:
forward: yes
masquerade: no
forward-ports:
source-ports:
icmp-blocks:
rich rules:
[root@prometheus prometheus]#
[root@prometheus prometheus]# ss -antlpu |grep 9090
tcp LISTEN 0      4096             *:9090           *:*               users:(("prometheus",pid=4881,fd=7))
```

Step 15:- Verify it's working

Open any browser and type below url

<http://<your-server-ip>:9090>



You should see the Prometheus web UI.

Navigate to Status → Target health and confirm you see one target listed

localhost:9090 | State: UP

That UP means Prometheus is successfully scraping its

own internal metrics — the installation is complete and working correctly.

Now, Download Alertmanager on Prometheus Server

Step 1: Download Alertmanager

```
#cd /opt

#wget
https://github.com/prometheus/alertmanager/releases/download/v0.33.1/alertmanag
er-0.33.1.linux-amd64.tar.gz

#ls
```

Step 2: Extract

```
#tar -xvzf alertmanager-0.33.1.linux-amd64.tar.gz
#ls
#rm -rf alertmanager-0.33.1.linux-amd64.tar.gz
#ls
#mv alertmanager-0.33.1.linux-amd64/ alertmanager
#ls
```

```

[root@prometheus opt]# wget https://github.com/prometheus/alertmanager/releases/download/v0.33.1/alertmanager-0.33.1.linux-amd64.tar.gz
--2026-07-14 23:30:09-- https://github.com/prometheus/alertmanager/releases/download/v0.33.1/alertmanager-0.33.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.42
Connecting to github.com (github.com):20.207.73.42... connected.
HTTP request sent, awaiting response... 302 Found
Location: /releases/assets/githubusercontent.com/github-production-releases/assets/14105338/1410458-56dc-43b3-b6-31d6f4cffe9f?raw=true [100%]
[100%] alertmanager-0.33.1.linux-amd64.tar.gz saved [38309254/38309254]
[100%] alertmanager-0.33.1.linux-amd64.tar.gz saved [38309254/38309254]
Connecting to github.com (github.com):20.207.73.42... connected.
HTTP request sent, awaiting response... 200 OK
Length: 38309254 (37M) [application/octet-stream]
Saving to: 'alertmanager-0.33.1.linux-amd64.tar.gz'

alertmanager-0.33.1.linux-amd64 100%[=====] 36.53M 7.10MB/s in 4.7s

2026-07-14 23:30:14 (7.82 MB/s) - 'alertmanager-0.33.1.linux-amd64.tar.gz' saved [38309254/38309254]

[root@prometheus opt]# tar -xvzf alertmanager-0.33.1.linux-amd64.tar.gz
alertmanager-0.33.1.linux-amd64/
alertmanager-0.33.1.linux-amd64/amtol
alertmanager-0.33.1.linux-amd64/LICENSE
alertmanager-0.33.1.linux-amd64/alertmanager
alertmanager-0.33.1.linux-amd64/NOTICE
alertmanager-0.33.1.linux-amd64/alertmanager.yml
[root@prometheus opt]#
[root@prometheus opt]#
[root@prometheus opt]#
[root@prometheus opt]# ls
alertmanager-0.33.1.linux-amd64.tar.gz prometheus
[root@prometheus opt]#
[root@prometheus opt]#
[root@prometheus opt]# mv -rf alertmanager-0.33.1.linux-amd64.tar.gz
removed 'alertmanager-0.33.1.linux-amd64.tar.gz'
[root@prometheus opt]#
[root@prometheus opt]# mv alertmanager-0.33.1.linux-amd64/ alertmanager
[root@prometheus opt]#
[root@prometheus opt]# ls

```

Step 3: Create User

```
#useradd --no-create-home --shell /bin/false alertmanager
```

Step 4: Change Ownership

```
#chown -R alertmanager:alertmanager /opt/alertmanager
```

Step 5: Create Data Directory

```
#mkdir -p /var/lib/alertmanager
```

```
#chown alertmanager:alertmanager /var/lib/alertmanager
```

Step 6: Create Configuration

```
#mkdir -p /etc/alertmanager
```

```
[root@prometheus opt]# useradd --no-create-home --shell /bin/false alertmanager
[root@prometheus opt]#
[root@prometheus opt]# chown -R alertmanager:alertmanager /opt/alertmanager
[root@prometheus opt]#
[root@prometheus opt]# mkdir -p /var/lib/alertmanager
[root@prometheus opt]#
[root@prometheus opt]# chown alertmanager:alertmanager /var/lib/alertmanager
[root@prometheus opt]#
[root@prometheus opt]# mkdir -p /etc/alertmanager
[root@prometheus opt]#
```

Create the configuration file:

```
#vi /etc/alertmanager/alertmanager.yml
```

global:

```
smtp_smarthost: 'smtp.gmail.com:587'
smtp_from: 'dineshcpatel0609@gmail.com'
smtp_auth_username: 'dineshcpatel0609@gmail.com'
smtp_auth_password: 'YOUR_GMAIL_APP_PASSWORD'
smtp_require_tls: true
```

route:

```
receiver: 'email-notifications'
group_wait: 10s
group_interval: 30s
repeat_interval: 1h
```

receivers:

```
- name: 'email-notifications'
  email_configs:
    - to: 'dineshcpatel0609@gmail.com'
```

send_resolved: true

:wq!

Note:- Replace YOUR_GMAIL_APP_PASSWORD with your Gmail App Password (not your normal Gmail password).

Step 7: Create Service

```
#vi /etc/systemd/system/alertmanager.service
```

[Unit]

Description=Prometheus Alertmanager

After=network.target

[Service]

User=alertmanager

Group=alertmanager

ExecStart=/opt/alertmanager/alertmanager \

--config.file=/etc/alertmanager/alertmanager.yml \

--storage.path=/var/lib/alertmanager

Restart=always

[Install]

WantedBy=multi-user.target

Step 8: Reload Systemd

```
#systemctl daemon-reload
```

Step 9: Enable Alertmanager

```
#systemctl enable --now alertmanager
```

Step 10: Start Alertmanager

```
#systemctl start alertmanager
```

Step 11: Check Status

```
#systemctl status alertmanager
```

```
[root@prometheus opt]# systemctl daemon-reload
[root@prometheus opt]#
[root@prometheus opt]# systemctl enable --now alertmanager.service
Created symlink /etc/systemd/system/multi-user.target.wants/alertmanager.service → /etc/systemd/system/alertmanager.service.
[root@prometheus opt]#
[root@prometheus opt]# systemctl status alertmanager.service
● alertmanager.service - Prometheus Alertmanager
   Loaded: loaded (/etc/systemd/system/alertmanager.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 23:35:49 IST; 16s ago
     Main PID: 5390 (alertmanager)
        Tasks: 8 (limit: 10301)
       Memory: 12.7M (peak: 13.0M)
          CPU: 50ms
      CGroup: /system.slice/alertmanager.service
              └─5390 /opt/alertmanager/alertmanager --config.file=/etc/alertmanager/alertmanager.yml --storage.path=/var/lib/a
```

Step 12: Open Firewall

```
#firewall-cmd --permanent --add-port=9093/tcp
```

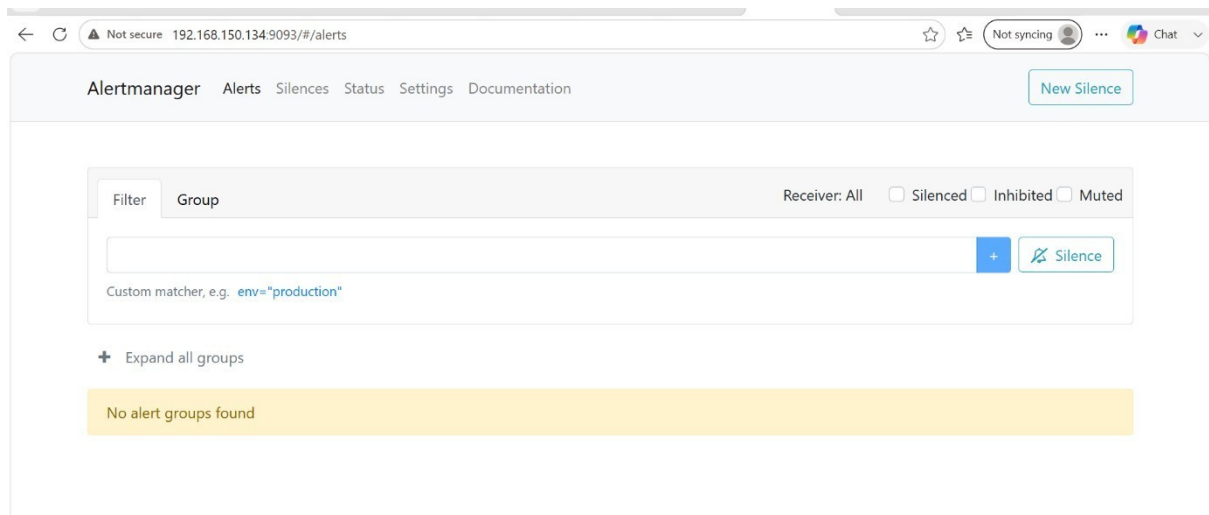
```
#firewall-cmd --reload
```

```
#firewall-cmd --list-all
```

```
[root@prometheus opt]# firewall-cmd --permanent --add-port=9093/tcp
success
[root@prometheus opt]#
[root@prometheus opt]# firewall-cmd --reload
success
[root@prometheus opt]#
[root@prometheus opt]# firewall-cmd --list-all
public (active)
  target: default
 icmp-block-inversion: no
 interfaces: ens160
  sources:
 services: cockpit dhcpv6-client ssh
 ports: 8080/tcp 9090/tcp 9093/tcp
 protocols:
 forward: yes
 masquerade: no
 forward-ports:
 source-ports:
 icmp-blocks:
 rich rules:
```

Step 13: Verify

```
http://<prometheus server-ip>:9093
```



Now Configure Prometheus

Step 14:- Edit the Prometheus configuration file.

```
#vi /etc/prometheus/prometheus.yml
```

remove existing line & add below line

global:

```
scrape_interval: 15s
```

rule_files:

```
- "/etc/prometheus/alert.rules.yml"
```

alerting:

alertmanagers:

```
- static_configs:
```

```
- targets:
```

```
- "192.168.150.134:9093"
```

scrape_configs:

- job_name: 'prometheus'

- static_configs:

- targets:

- 'localhost:9090'

- job_name: 'jenkins'

- metrics_path: /prometheus

- static_configs:

- targets:

- '192.168.150.129:8080'

- job_name: 'node'

- static_configs:

- targets:

- '192.168.150.129:9100'

- job_name: 'apache'

- static_configs:

- targets:

- '192.168.150.129:9117'

:wq!

#systemctl restart Prometheus

```
[root@prometheus opt]# systemctl restart prometheus.service
```

Step 15:-Verify Prometheus Targets

<http://<prometheus server-ip>:9090/targets>

Prometheus			
Query Alerts Status > Target health			
Not secure 192.168.150.134:9090/targets			
0 / 1 up			
Endpoint	Labels	Last scrape	State
http://192.168.150.129:9117/metrics	instance="192.168.150.129:9117" job="apache"	never 0s	UNKNOWN
jenkins			
0 / 1 up			
Endpoint	Labels	Last scrape	State
http://192.168.150.129:8080/prometheus	instance="192.168.150.129:8080" job="jenkins"	never 0s	UNKNOWN
node			
1 / 1 up			
Endpoint	Labels	Last scrape	State
http://192.168.150.129:9100/metrics	instance="192.168.150.129:9100" job="node"	543ms ago 45ms	UP
prometheus			
0 / 1 up			
Endpoint	Labels	Last scrape	State
http://localhost:9090/metrics	instance="localhost:9090" job="prometheus"	never 0s	UNKNOWN

Verify that the following targets are in the UP state.

Job	Status
-----	-----
Prometheus	UP
Jenkins	UP
Node Exporter	UP
Apache Exporter	UP

Step 16:- Create Alert Rules

```
#vi /etc/prometheus/alert.rules.yml
```

groups:

- name: MonitoringAlerts

rules:

- alert: HighCPUUsage

expr: 100 - (avg by(instance)(rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100) > 80

for: 5m

labels:

severity: critical

annotations:

summary: "High CPU Utilization"

description: "CPU utilization is above 80%."

- alert: ApacheServiceDown

expr: up{job="apache"} == 0

for: 1m

labels:

severity: critical

annotations:

summary: "Apache Service Down"

description: "Apache service is not reachable."

- alert: JenkinsBuildFailed

expr: default_jenkins_builds_last_build_result_ordinal == 2

for: 1m

labels:

severity: critical

annotations:

summary: "Jenkins Build Failed"

description: "The latest Jenkins build has failed."

:wq!

Restart Prometheus.

```
#systemctl restart prometheus
```

```
#systemctl status prometheus
```

Step 17: Configure Grafana

Log in to Grafana.

<http://<grafana server-ip>:3000>

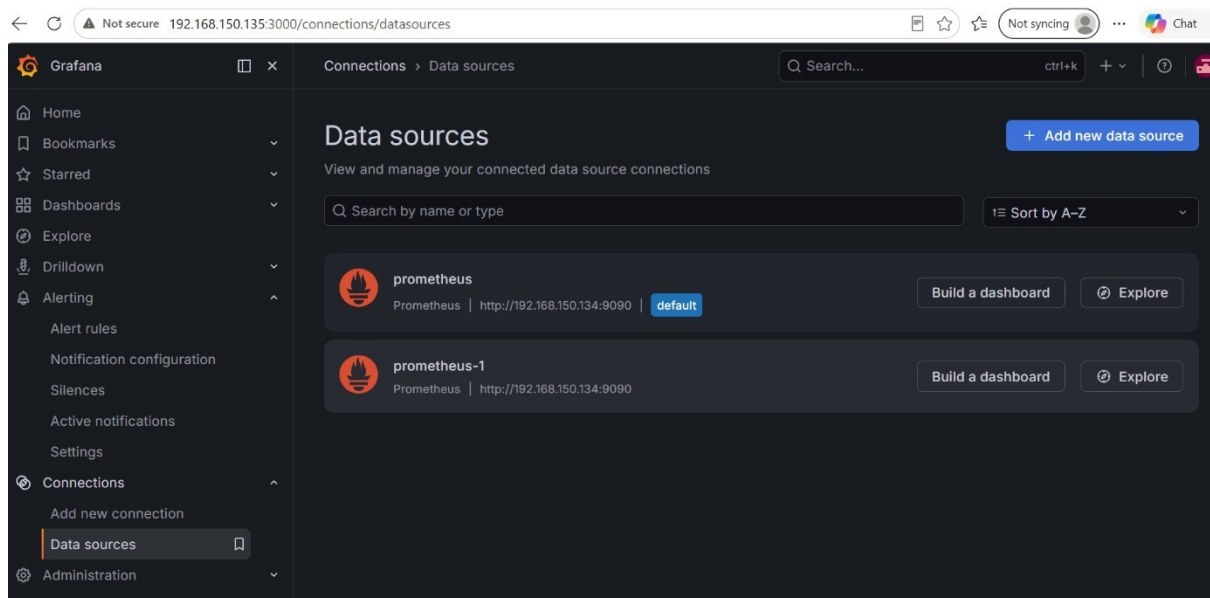
Navigate to:

Connections

→ Data Sources

→ Add Data Source

→ Prometheus



Configure the Prometheus URL:

<http://<prometheus server-ip>:9090>

Click Save & Test.

Expected message:

Data source is working.

Step 18: Create Grafana Dashboard

Create a new dashboard and add the following panels.

Panel 1: CPU Utilization

promql

```
100 - (avg by(instance)(rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100)
```

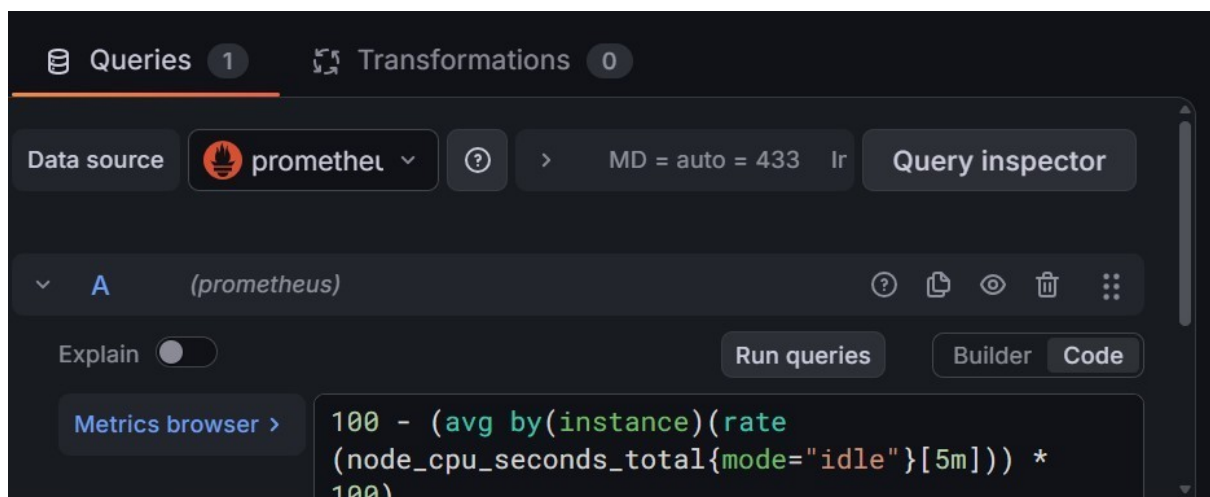
run queries

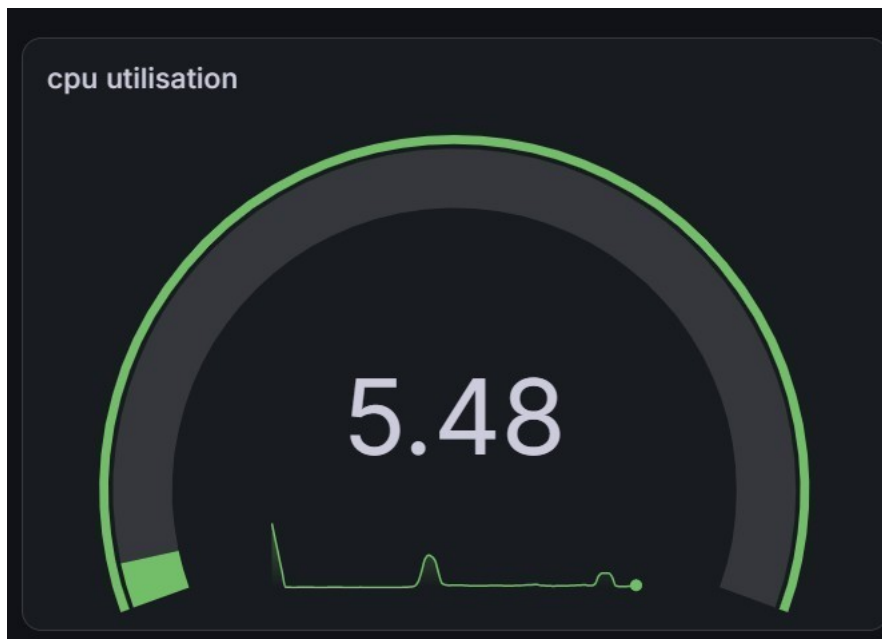
Visualization:

Gauge

Panel option

inside Title: CPU Utilization and save the dashboard





Now add Panel 2: Memory Utilization

promql

$100 * (1 - (\text{node_memory_MemAvailable_bytes} / \text{node_memory_MemTotal_bytes}))$

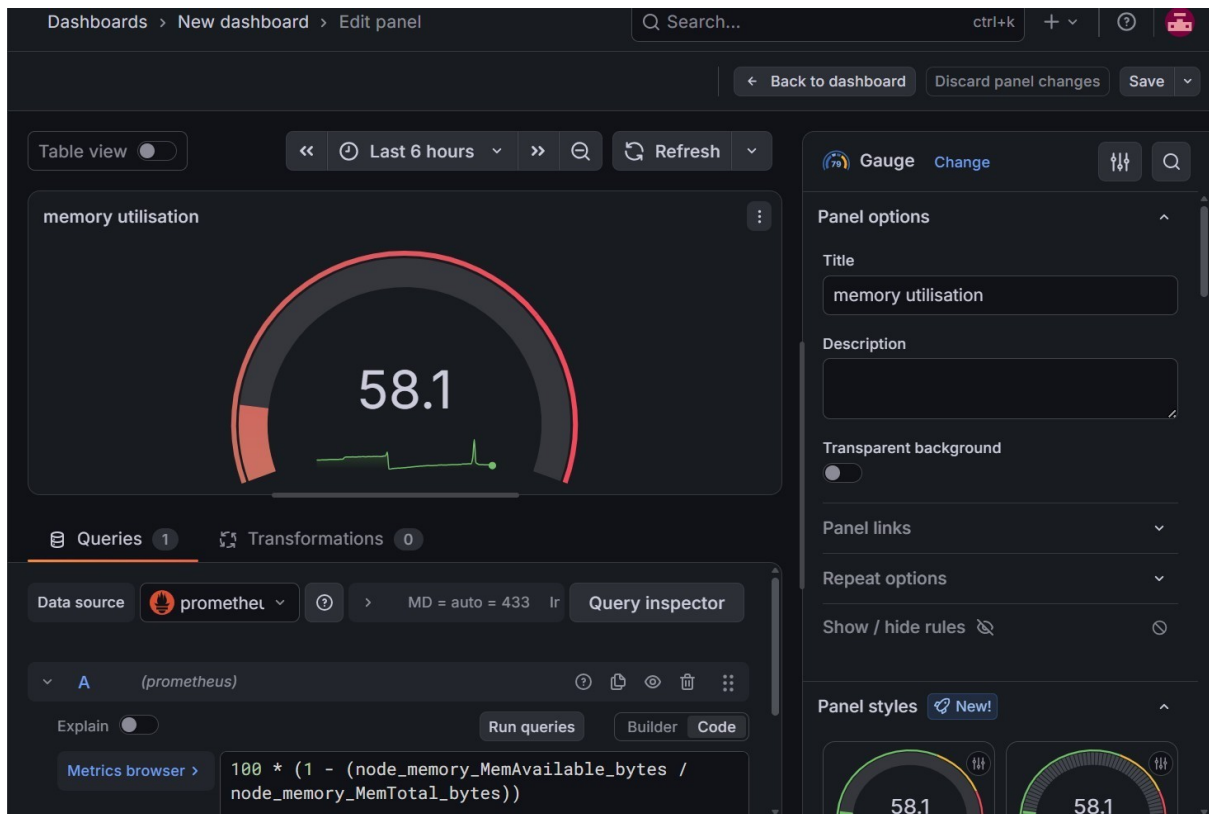
run queries

Visualization:

Gauge

Panel option

inside Title:Memory Utilization and save the dashboard



Panel 3: Apache Service Status

promql

apache_up

run queries

Visualization:

Stat

scroll down

Click + Add value mapping.

| Value | Display text | Color |

| ---- | ----- | ---- |

| `1` | `UP` | Green |

| `0` | `DOWN` | Red |

save with name Apache Service Status

Value mappings
✕

Condition	Display text	Color
Value 1	up	● ✕
Value 0	down	● ✕

+ Add a new mapping
Cancel
Update

Dashboards > New dashboard > Edit panel
Q Search...
ctrl+k
+
?

← Back to dashboard
Discard panel changes
Save

Table view
⏮
⌚ Last 6 hours
⏭
🔍
🔄 Refresh

Apache service status

up

Queries 1

Transformations 0

Data source prometheus

MD = auto = 433

Query inspector

A (prometheus)

Explain

Run queries

Builder

Code

Metrics browser >

apache_up

12.4 Stat Change

Panel options

Title

Apache service status

Description

Transparent background

Panel links

Repeat options

Show / hide rules

Panel styles

New!

1

1

Panel 4: Jenkins Build Status

promql

default_jenkins_builds_last_build_result_ordinal

jenkins_plugins_failed

run queries

Visualization:

Stat

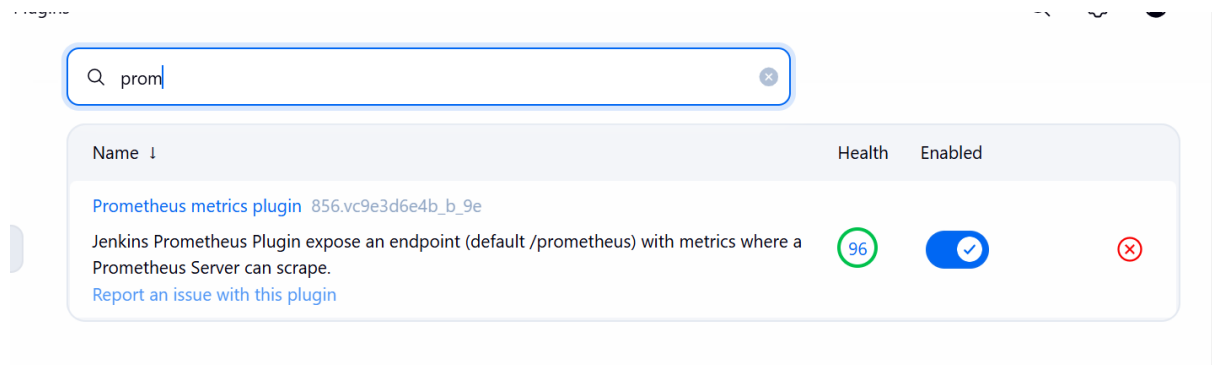
scroll down

Click + Add value mapping.

Value | Display text | Color |

```
| ---- | ----- | ---- |
| `1` | `Success` | Green |
| `0` | `Failed` | Red
```

save with name Jenkins Build Status



Panel 5: Disk Utilization (%)

PromQL Query

```
100 - (
  (node_filesystem_avail_bytes{mountpoint="/",fstype!~"tmpfs|overlay"}
    * 100)
  /
  node_filesystem_size_bytes{mountpoint="/",fstype!~"tmpfs|overlay"})
```

)

This query shows the percentage of disk space used on the root (/) filesystem.

Visualization

Type: Gauge

Unit: Percent (0-100)

Min: 0

Max: 100

Thresholds

Configure the thresholds as follows:

Color	Value
-------	-------

-----	----
-------	------

 Green	0
---	---

 Yellow	70
--	----

 Red	90
---	----

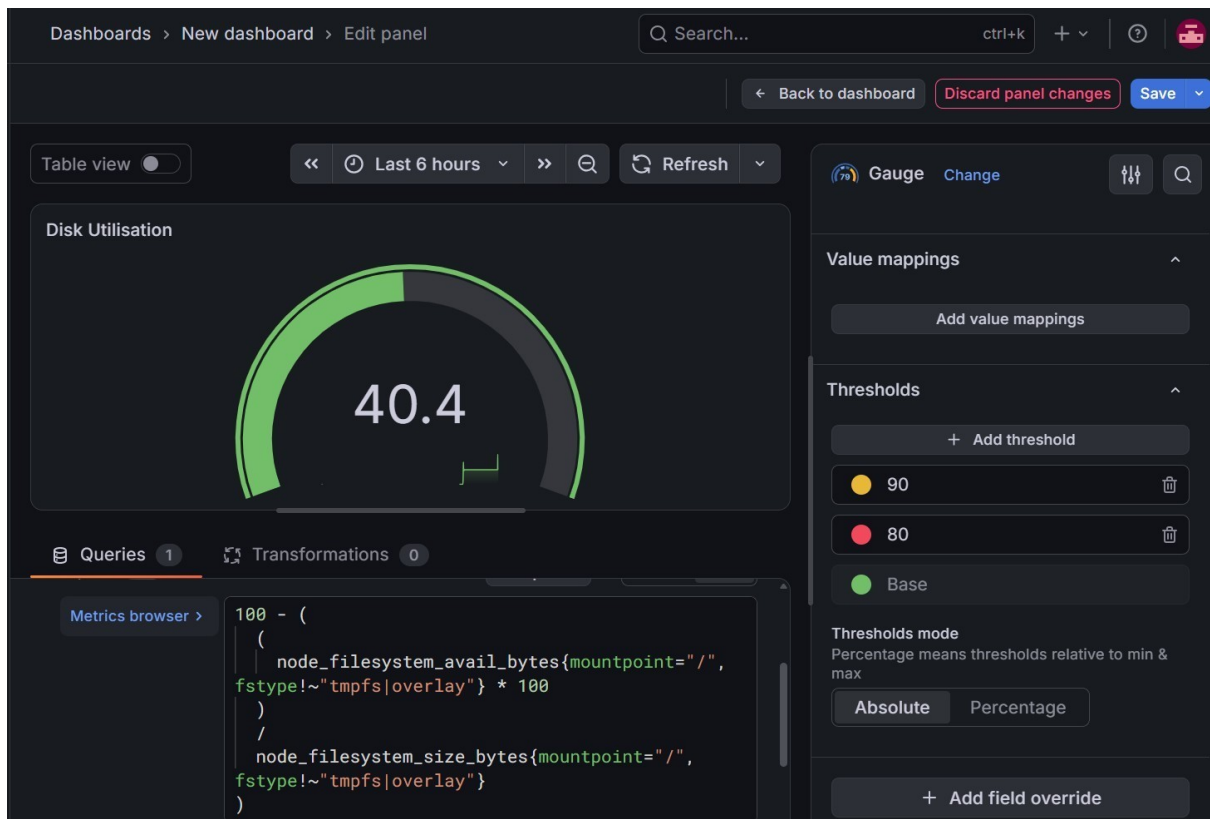
This means:

0–70% → Green (Healthy)

70–90% → Yellow (Warning)

90–100% → Red (Critical)

save with name Running Disk Utilization (%)



Now Test the Alerts on jenkins server

so go back to jenkins server perform the steps



Step 10: Test the Alerts

Test High CPU Utilization

Install the stress utility.

install epel because stress package is available in epel repo

```
#yum install -y https://dl.fedoraproject.org/pub/epel/epel-release-latest-9.noarch.rpm
```

```
#yum install stress -y
```

Generate CPU load.

```
#stress --cpu 4 --timeout 300
```

```
root@localhost opt]# stress --cpu 4 --timeout 300  
stress: info: [19981] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
```

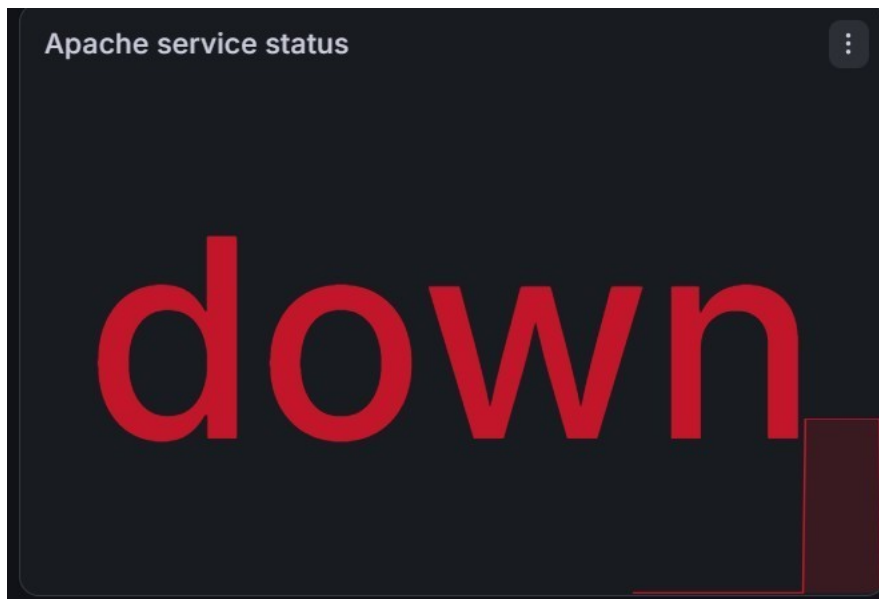
go to grafana server and check cpu utilization



Step 11:- Test Apache Service Down

```
#yum install httpd -y
```

```
#systemctl stop httpd
```



```
#systemctl start httpd
```

Step 12:- now go to jenkins dashboard create one job

job name:-My_First_Maven_Project

git:- to paste github account FormFillApp repo url

branch:-main

Goals and options:-clean & install {this is wrong command to failed the jenkins job}

apply and save and build a job

so we can easily monitor jenkins failed alerts in prometheus and grafana.

Step 13:- to Monitor with email

Generate a Gmail App Password

Since you're using Gmail (dineshcpatel0609@gmail.com), you cannot use your normal Gmail password.

to set app password

<https://myaccount.google.com/apppasswords>

App name:-Alertmanager

Click Create.

Google provide 16-character App Password

Step 14:- On the Prometheus Server (192.168.58.133)

open below file and paster apppasswords to this file

```
# vi /etc/alertmanager/alertmanager.yml
```

Replace:

YOUR_GMAIL_APP_PASSWORD

Step 15:- Validate the Configuration

```
/opt/alertmanager/amtool check-config /etc/alertmanager/alertmanager.yml
```

Expected output:

SUCCESS

Step 16:- Restart Alertmanager & prometheus

```
#systemctl restart alertmanager
```

```
#systemctl restart prometheus
```

Step 17:- To Integrate Jenkins Success and Failed Job ,CPU,Memeory,Apache, disk utilization alerts

via Grafana email

Now go back to grafana server

and open the below file

```
#vim /etc/grafana/grafana.ini
```

go to last line

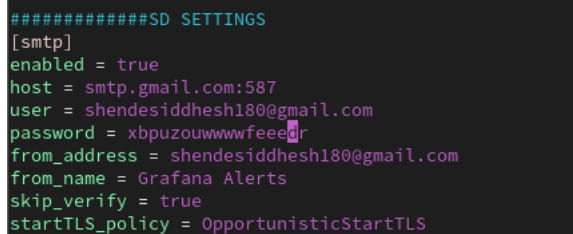
make sure demarktession for this line

```
#####below settings done by dinesh  
patel#####
```

```
[smtp]
```

```
enabled = true
host = smtp.gmail.com:587
user = dineshcpatel0609@gmail.com
password = YOUR_GMAIL_APP_PASSWORD
from_address = dineshcpatel0609@gmail.com
from_name = Grafana Alerts
skip_verify = true
startTLS_policy = OpportunisticStartTLS
```

:wq! ----->{force fully save and quite}



```
#####SD SETTINGS
[smtp]
enabled = true
host = smtp.gmail.com:587
user = shendesiddhesh180@gmail.com
password = xbpuzouwwwfееееr
from_address = shendesiddhesh180@gmail.com
from_name = Grafana Alerts
skip_verify = true
startTLS_policy = OpportunisticStartTLS
```

Now then restart grafana service

```
#systemctl restart grafana-server
```

Step 18:- Now go to grafana dashboard

go to Alerting

inside Alert rules

create new alert rule

i)Enter alert rule name:- Grafana Alert

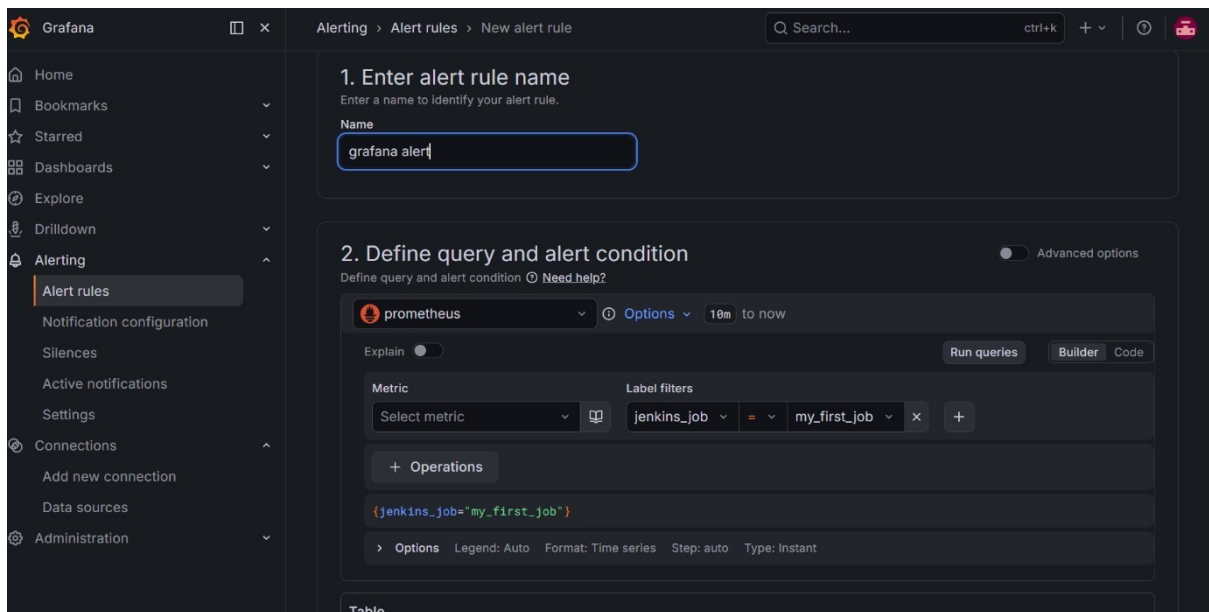
ii) Define query and alert condition

Metric:-

Label filters:- jenkins_job

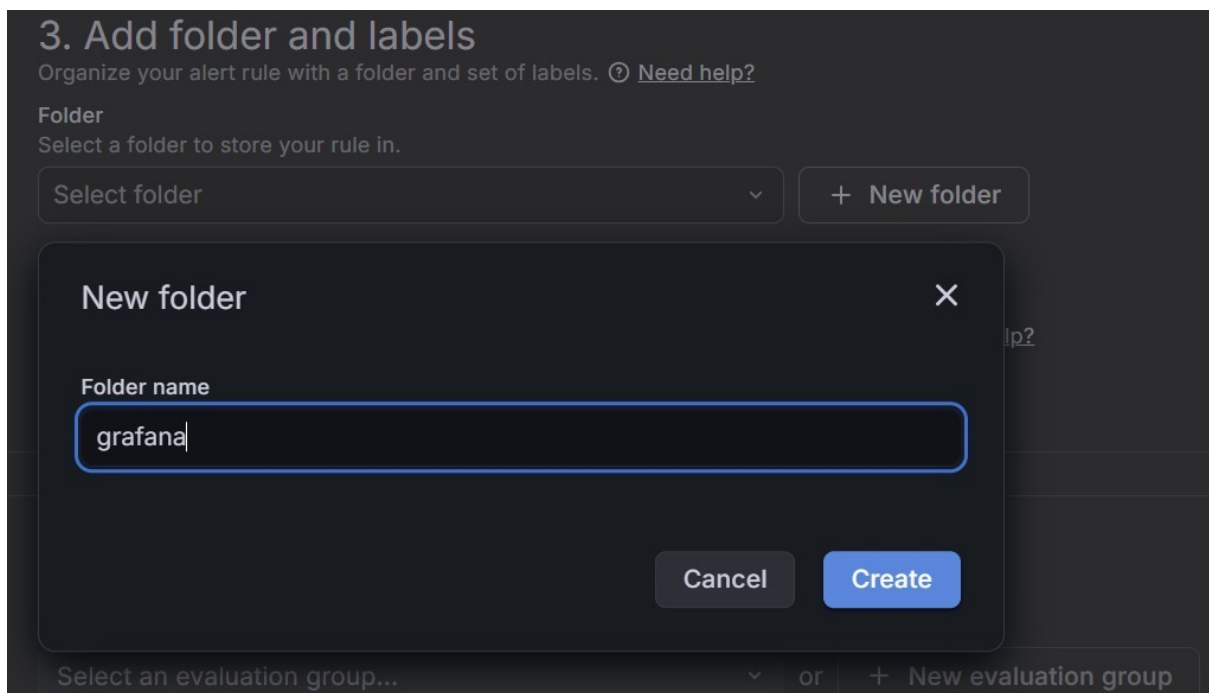
Select value:- My_First_Job

Run queries



iii) Add folder and labels

New folder:- Grafana



iv) Set evaluation behavior

add evaluation group

name:- grafana-Alerts

evaluation_interval:- 1m

and then create

New evaluation group ✕

Create a new evaluation group to use for this alert rule.

Evaluation group name
A group evaluates all its rules over the same evaluation interval.

grafana-alert

Evaluation interval
How often all rules in the group are evaluated.

1m

10s 30s **1m** 5m 10m 15m 30m 1h

Cancel Create

v) Configure notifications

click on View or create contact points

then new tab is open

click on new contact points

Name:- Jenkins_Job_Alerts_Latest

Integratation:- Email

Address:- your personal email id

and then click on Test

Notification Message:- Custom

Summary:- Jenkins_Job_Failed

and send notification

Create contact point

Name *

Apache-down

Integration

Email

Addresses

You can enter multiple email addresses using a ";", "\n" or "," separator

shendesiddhesh180@gmail.com

> Optional Email settings

> Notification settings

Test Duplicate Delete

and then save contact point

Step 10:- Now check your personal email id

you will receive TestAlert Grafana

it will see Jenkins_Job_Failed

alerts

 1 firing instances

Firing		DatasourceNoData	View alert
Summary			
apache service down			
Description			
apache service down			
Labels			
alertname	DatasourceNoData		
datasource_uid	ffs3yowyjueiod		
grafana_folder	grafana		
ref_id	A		
rulename	grafana alert		